

```
$ ./presentation.sh
```

```
// ML-accelerated IR spectroscopy
```

\$ cat motivation.md

Problem

- DFT frequency calculations: slow
- Minutes to hours per molecule

Solution

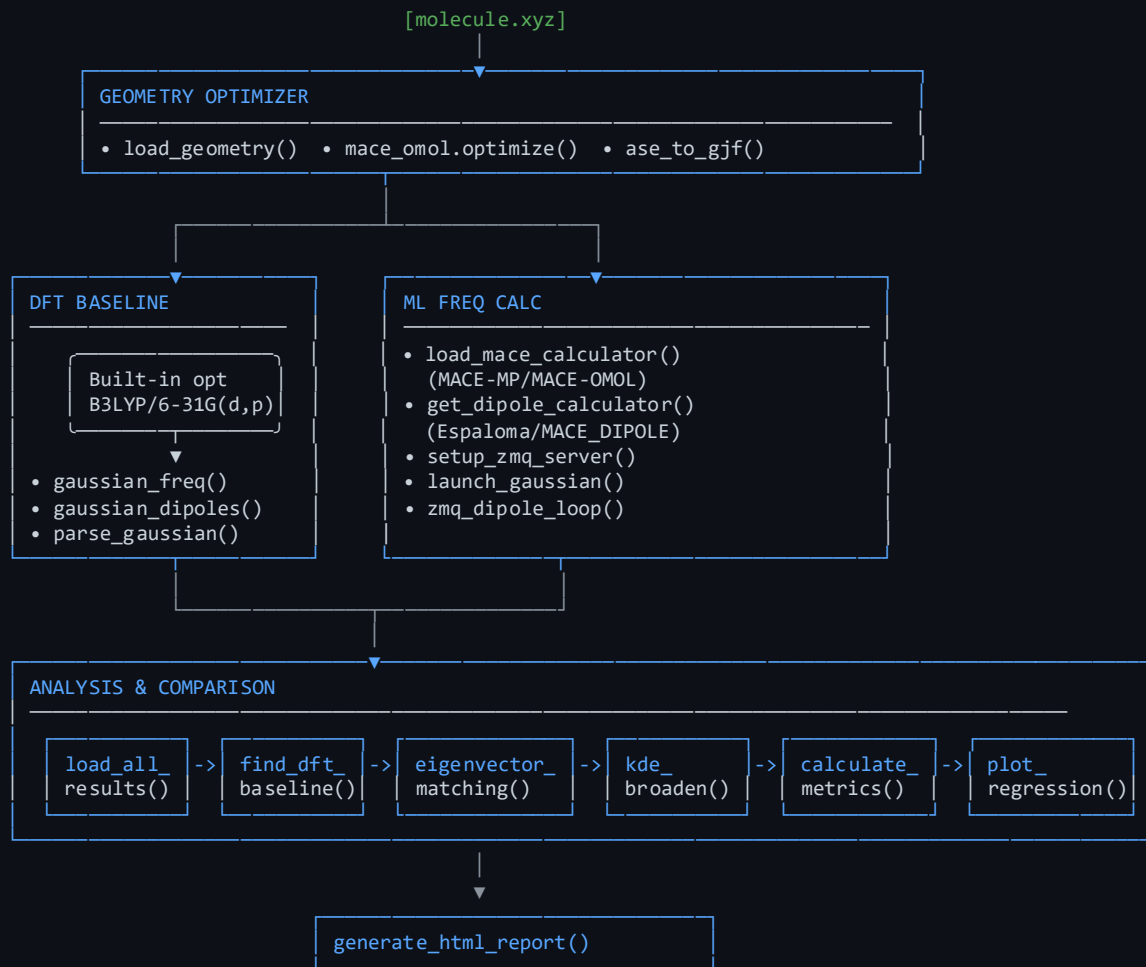
- Hybrid ML-QM approach
- ML dipoles and energies
- Possibly 10-100× faster than pure DFT

Impact

- Enable rapid IR spectral predictions
- Maintain DFT-level accuracy

\$ python3 main.py

System Architecture



\$ ls -l calculators/

```
# Modular Dipole System
```

```
mace_dipole/      # Custom MACE
```

```
espaloma/         # ML charges
```

```
# Implementation
```

- Factory pattern with availability checks and warnings
- Seamless calculator switching
- Automatic combination with available energy calculators

\$ cat dipole_methods.md

MACE-Dipole

Input $\rightarrow$ E(3)-equivariant NN $\rightarrow \mu = (\mu_x, \mu_y, \mu_z)$

$\rightarrow$ Direct dipole prediction

$\rightarrow$ Fast

$\rightarrow$ Requires custom model

Espaloma (Charge-Based)

Predict charges $\rightarrow \mu = \sum q_i \cdot r_i$

$\rightarrow$ ML partial charges ($q_1, q_2, \dots, q_n$)

$\rightarrow$ Sum over atomic positions

$\rightarrow$ Available as package

\$ python3 workflow_manager.py

Automatic Directory Structure

```
comparison_results/  
└─ {molecule}/  
    ├── geometry_opt/  
    ├── b3lyp/  
    └─ {energy_calc}_{dipole_calc}/
```

- Created on-the-fly
- Organized by calculator combination

Smart File Detection

- Auto-detect existing calculations
- Skip completed steps
- Find DFT baseline automatically

Results Aggregation

- Load all JSON results
- Convert .chk to .fchk
- Parse all .log and .fchk files
- Match modes across methods
- Save tables and plots
- Generate unified HTML report

\$ ls -la parsing/

Gaussian .log Files (2-10 MB)

- Harmonic frequencies
- Anharmonic frequencies
- Overtones & combination bands
- IR intensities
- Eigenvectors (3N-6 modes)

Challenges:

- Multi-column format (3 modes/block)
- 10,000-50,000 lines to search

JSON Results

```
{  
  "frequencies": [...],  
  "ir_intensities": [...],  
  "eigenvectors": [[...]]  
}
```

→ Structured, easy to analyze

\$ cat mode_matching.md

Problem: Frequency Order Mismatch

DFT: [500, 1200, 1500, 2900, 3100] cm^{-1}

ML: [498, 3095, 1205, 2895, 1498] cm^{-1}

→ Cannot compare by index

Solution: Eigenvector Dot Product

similarity = $|\mathbf{v}_{\text{ML}} \cdot \mathbf{v}_{\text{DFT}}|$

→ Compare atomic displacement patterns

→ Find best match

→ Robust to frequency ordering

Example

ML 3095 cm^{-1} → DFT 1200 cm^{-1} [0.027]

ML 3095 cm^{-1} → DFT 3100 cm^{-1} [0.989]


```
$ cd results/
```

```
# Summary in Form of HTML Report
```

```
water.html
```

\$ git log --oneline --graph

Current Status

- ZMQ bridge implemented & working
- Small molecule benchmarks validation in progress
- Solid automated workflow orchestration in place
- Comparison of methods in one place

Next Steps

- Expand to larger molecules
- Anharmonic frequency analysis
- Command line interface
- ORCA support (open source)

```
$ ./questions.sh
```